Manan Sawrangpate

manan.sawrangpate@mail.utoronto.ca
Github ▪ LinkedIn

Education

University of Toronto - BASc, Engineering Science (Major in **Robotics Engineering**) 2022 – 2027 (expected)
Minors: Engineering Business (Rotman) | **Awards:** Alexander Rutherford ($2.5k), Wallberg ($5k)

Experience

Embedded Software Engineering Intern – Hedgehog Technologies – Burnaby, BC May 2026 – Present

- Developed embedded C firmware for a virtual EEPROM on the NXP S32K144 using internal flash peripheral drivers, implementing Hardware Abstraction Layers that enabled the team to remove the physical EEPROM from the design.
- Brought up 2 FOC motor controller boards by characterizing 3-phase SVPWM waveforms (0–4000/24V & 0–7500 RPM /48V) via oscilloscope measurement, developing a validation test plan to confirm motor drive signal integrity.
- Diagnosed repeated board faults to a MOSI-CLK short on the motor driver IC by probing the full SPI bus (MOSI, MISO, CS, CLK) and resolving via component replacement to restore nominal FOC operation.
- Identified and resolved bring-up failure modes (bootstrap capacitor sizing, failed MOSFETs) across 3 boards by tracing faults through Altium schematics; documented findings and component modifications to maintain accurate board status

Process Engineering Intern – York Region – Newmarket, ON May 2025 – April 2026

- Engineered a two-stage Python classifier (keyword filter + TF-IDF/Complement Naive Bayes) to triage 4,000–5,000 monthly operator logs, achieving 100% issue recall on validation data & reducing manual review volume by 55%.
- Derived geometric volume models for septage tanks from structural drawings, developing a linearized approximation in Excel suitable for SCADA integration and delivering live operational KPIs to ±0.1 m^3 accuracy across pump stations.
- Built Excel pump cycle models from indirect operational data, informing EQ tank sizing decisions & saving $18K annually.
- Produced 5+ P&IDs and process schematics in Bluebeam and Visio, incorporating As-Built revisions from onsite inspections to support facility-wide deployment across York Region facilities.

Selected Projects

Autonomous Mail Delivery Robot with Bayesian Localization (GitHub) Nov 2024 – Dec 2024

- Developed a ROS2 Jazzy autonomous mobile robot system on Linux (Ubuntu), using a multi-threaded executor for concurrent perception & control nodes, integrating a PID line-following controller driven by RGB camera feedback.
- Designed a recursive Bayesian localization filter with a 12-state transition model (P=0.85 advance) and 5-class colour measurement model to resolve positional ambiguity across a closed-loop 12-state map, achieving >90% state confidence within 5 observations per trial; validated across 5 physical TurtleBot3 hardware trials, extended to Gazebo Harmonic.

Dynamic Stability Cart for Rope-Bridge Traversal (Report) Feb 2025 – Apr 2025

- Designed an autonomous balancing cart for rope bridge traversal using an STM32 Nucleo and 9-DoF IMU over SPI; cut control-loop latency by >95% via embedded C++ DMP sensor fusion and gain tuning on a PI-controlled servo rod.
- Validated stability through iterative hardware testing across dual- to single-anchor configurations, diagnosing Kd-induced oscillations and converging on a tuned PI controller (Kp=1.5, Ki=0.02) to achieve >80% crossing success rate.

Design Teams

Aerodynamics Lead - *University of Toronto Wind Turbine Team (UTWind)* Sept 2023 – June 2025

- Led a 10-member aerodynamics sub-team in iterative CFD optimization of VAWT blade profiles using ANSYS Fluent and Discovery, with results informing blade selection for 1st place at 2024 International Small Wind Turbine Competition.

Skills

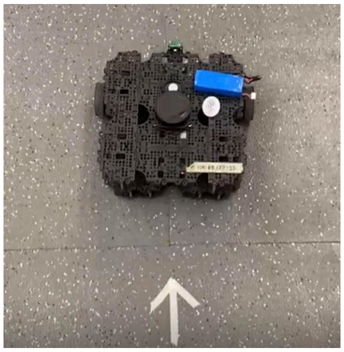
- **Software:** Python, C/C++, Bash, Simulink/MATLAB, Git/Github | **Protocols**: I2C, SPI, CAN, UART, USB (2.0)
- **Robotics**: ROS2 (Jazzy), ROS1, Motion Planning, Autonomous Navigation, Multithreading, Embedded Systems
- **Hardware:** Oscilloscope, Soldering, Signal Probing, Multimeter, Electrical Assembly, Software-Hardware Interfacing
- **Technologies/Tools:** ARM (STM32, NXP), Linux (Ubuntu ROS1/2), Eagle PCB, Altium, PSpice, Visual Studio, S32k IDE

Portfolio - Manan Sawrangpate

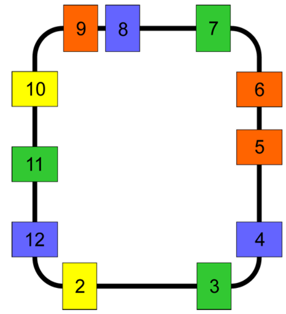
manan.sawrangpate@mail.utoronto.ca
Github ▪ LinkedIn

Autonomous Mail Delivery Robot

Final Report | Demo Video | Github



TurtleBot3 Waffle Pi Autonomous Delivery Robot

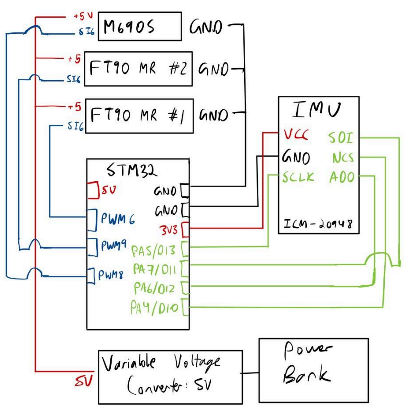


12-State Office Navigation Map

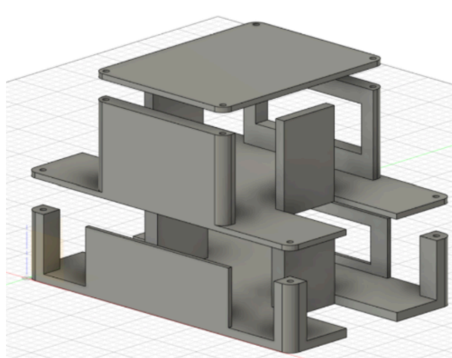
- Extended the physical ROS1 TurtleBot3 implementation to **ROS2** Jazzy / **Gazebo** Harmonic, recalibrating colour classification thresholds from real camera to Ogre2 renderer while preserving the 12-state map and Bayesian filter.
- Applied a belief floor of 0.01 after every update cycle to prevent permanent state elimination, ensuring filter recovery from misclassified colour observations without irreversible state collapse.
- Implemented a corridor-freeze strategy suppressing filter prediction during mid-corridor traversal and firing exactly once on patch entry, preventing belief double-advancement between offices.
- Architected a two-phase state machine (exploration lap to delivery), gating delivery on 8 consecutive MAP-confident frames to prevent premature triggering; achieved 60% full 3-office delivery completion across 5 physical trials.

Dynamic Stability Cart for Rope-Bridge Traversal

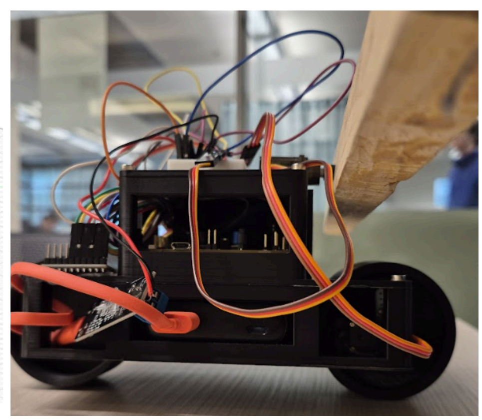
Final Report | Demo | Cart Overview



System Diagram (STM32 & connections)



Deconstructed view of Cart



Side View of Cart with Balancing Beam

- Engineered a two-wheeled autonomous cart to traverse a custom-built rope-suspended bridge, integrating an **STM32** NUCLEO-G070RB with a 9-DoF ICM-20948 IMU over **SPI** and a servo-actuated balancing rod; developed embedded C++ firmware implementing PI control for real-time pitch stabilization, achieving <3° steady-state error and consistent successful bridge crossings on a 1 m single-axis rail.
- Reduced control-loop latency by >95% (5s to <100 ms) by offloading quaternion-based sensor fusion to the IMU's onboard DMP, eliminating tip-over events caused by delayed pitch correction.
- Developed two **Fusion 360** design iterations of 3D-printed chassis, transitioning from a high-wall enclosed structure to an open multi-level layout that reduced mass while maintaining structural integrity; enforced chassis symmetry to center the CoM on the cart's longitudinal axis, simplifying balancing problem and improving disturbance rejection on 1-axis rail.
- Validated system stability through iterative hardware testing across progressively harder rail configurations (dual-anchor to single-anchor string attachment), diagnosing Kd-induced oscillations that caused tip-overs and simplifying to a tuned PI controller (Kp=1.5, Ki=0.02) to achieve repeatable autonomous bridge traversal.